PHY 211

Lecture 7 Notes

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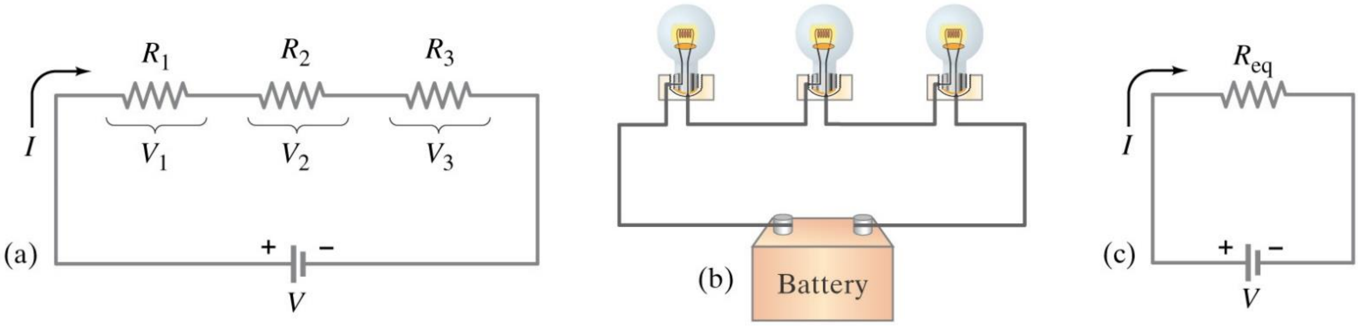
Fall 2024

Chapter 6
Direct Current "DC" Circuits

➢ Resistors in series & parallel:

- **Resistors in series:**



The same current I passes through R_1, R_2 & R_3.

The emf of the battery E is divided among the resistors

$$V_1 = IR_1 \qquad V_2 = IR_1 \qquad V_3 = IR_3$$

$$E = V_1 + V_2 + V_3$$

$$E = IR_1 + IR_1 + IR_3$$

$$E = I(R_1 + R_1 + R_3) \qquad (1)$$

The equivalent resistance R_{eq} of $(R_1, R_2 \,\&\, R_3)$ will draw the same current I

$$E = IR_{eq} \qquad (2)$$

From (1) & (2):

$$I(R_1 + R_1 + R_3) = IR_{eq}$$

$$\boxed{\boldsymbol{R_{eq} = R_1 + R_1 + R_3}}$$

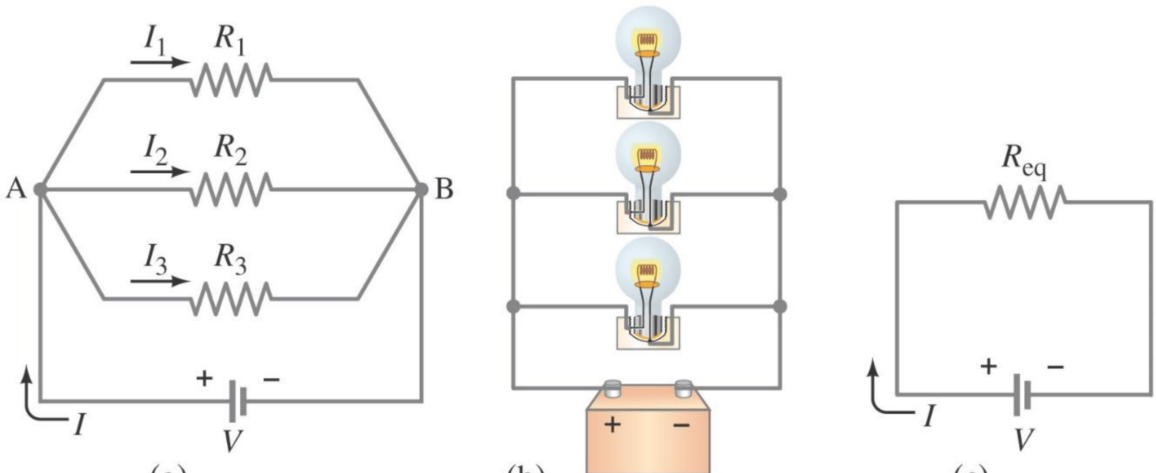
$$\boxed{\boldsymbol{R_{eq} > R_1 \quad \& \quad R_{eq} > R_2 \quad \& \quad R_{eq} > R_3}}$$

$$\boxed{\boldsymbol{\frac{V_1}{V_2} = \frac{R_1}{R_2} \quad \& \quad \frac{V_1}{E} = \frac{R_1}{R_{eq}}}}$$

For n resistors of same value R in series:

$$\boxed{\boldsymbol{R_{eq} = nR}}$$

- **Resistors in parallel:**



(a) (b) (c)

The voltage across R_1, R_2 & R_3 will be the same and equal to E.

The current I will be divided through R_1, R_2 & R_3 as I_1, I_2 & I_3

$$I_1 = \frac{E}{R_1} \qquad I_2 = \frac{E}{R_2} \qquad I_3 = \frac{E}{R_3}$$

$$I = I_1 + I_2 + I_3 \rightarrow I = \frac{E}{R_1} + \frac{E}{R_2} + \frac{E}{R_3}$$

$$I = E\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right) \qquad (1)$$

The equivalent resistance R_{eq} of $(R_1, R_2 \ \& \ R_3)$ will draw the same current I

$$I = \frac{E}{R_{eq}} \qquad (2)$$

From (1) & (2):

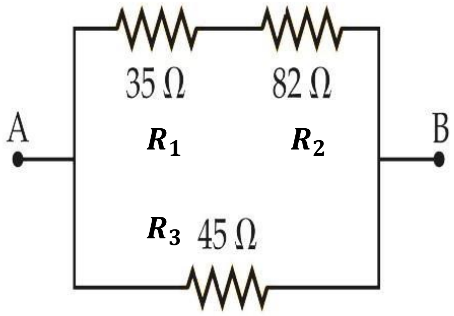
$$E\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right) = \frac{E}{R_{eq}}$$

$$\boxed{\frac{\mathbf{1}}{\boldsymbol{R_{eq}}} = \frac{\mathbf{1}}{\boldsymbol{R_1}} + \frac{\mathbf{1}}{\boldsymbol{R_2}} + \frac{\mathbf{1}}{\boldsymbol{R_3}}} \rightarrow \boxed{\boldsymbol{R_{eq} < R_1 \quad \& \quad R_{eq} < R_2 \quad \& \quad R_{eq} < R_3}}$$

$$\boxed{\frac{\boldsymbol{I_1}}{\boldsymbol{I_2}} = \frac{\boldsymbol{R_2}}{\boldsymbol{R_1}} \quad \& \quad \frac{\boldsymbol{I_1}}{\boldsymbol{I}} = \frac{\boldsymbol{R_{eq}}}{\boldsymbol{R_1}}}$$

For n resistors of same value R in parallel: $\boxed{\boldsymbol{R_{eq} = \frac{R}{n}}}$

- **<u>Example 1:</u>** Find the equivalent resistance between A and B for the group of resistors shown in figure below.



<u>Solution</u>

R_1 & R_2 are in series:

$$R_{12} = R_1 + R_2 = 35 + 82 = 117\ \Omega$$

R_{12} & R_3 are in parallel:

$$\frac{1}{R_{eq}} = \frac{1}{R_{12}} + \frac{1}{R_3}$$

$$R_{eq} = \frac{R_{12} \times R_3}{R_{12} + R_3} = \frac{117 \times 45}{117 + 45} = 32.5\ \Omega$$

- **<u>Example 2:</u>** A circuit consists of a 9.0-V battery connected to three resistors $(42\ \Omega, 17\ \Omega\ and\ 110\ \Omega)$ in series. Find

  (a) the current that flows through the battery and

  (b) the potential difference across each resistor.

<u>Solution</u>

$$E = 9\,V \qquad R_1 = 42\ \Omega \qquad R_1 = 17\ \Omega \qquad R_3 = 110\ \Omega$$

a)

$$I = \frac{E}{R_{eq}} = \frac{9}{42 + 17 + 110} = 0.053\ A = 53 \times 10^{-3}\ A = 53\ mA$$

b)

$$V_1 = IR_1 = 0.053\ \times 42 = 2.226\ V$$

$$V_2 = IR_2 = 0.053\ \times 17 = 0.901\ V$$

$$V_3 = IR_3 = 0.053\ \times 110 = 5.83\ V$$

- **<u>Example 3:</u>** A circuit consists of a battery connected to three resistors $65\ \Omega, 25\ \Omega\ and\ 170\ \Omega)$ in parallel. The total current through the resistors is 1.3 A. Find

  (a) the emf of the battery and

  (b) the current through each resistor.

<u>Solution</u>

$$I = 1.3\ A \qquad R_1 = 65\ \Omega \quad R_1 = 25\ \Omega \quad R_3 = 170\ \Omega$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{65} + \frac{1}{25} + \frac{1}{170} = 0.0612$$

$$R_{eq} = 16.34\ \Omega$$

a)

$$E = IR_{eq} = 1.3 \times 16.34 = 21.22\ V$$

b)

$$I_1 = \frac{E}{R_1} = \frac{21.22}{65} = 0.33\ A$$

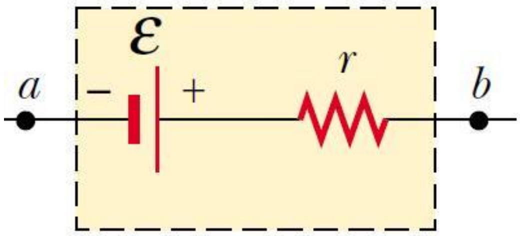
$$I_2 = \frac{E}{R_2} = \frac{21.22}{25} = 0.85\ A$$

$$I_3 = \frac{E}{R_3} = \frac{21.22}{170} = 0.12\ A$$

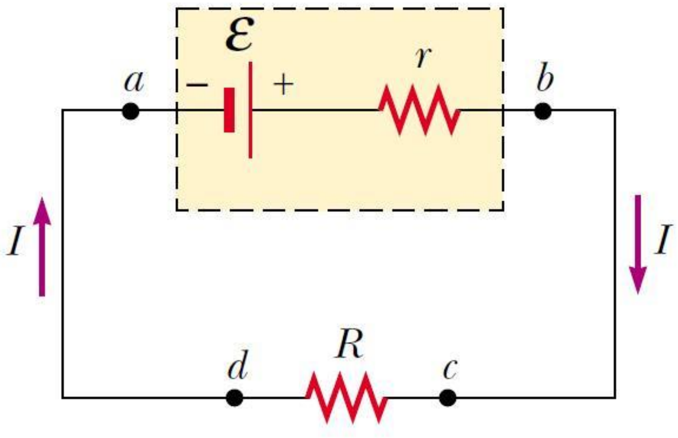
➢ **EMF & terminal voltage:**

- When no current flows to an external circuit (open circuit), the potential difference between the terminals of the source is called the emf (E)

$$V = \mathcal{E}$$



- The potential difference between the terminals of the battery (V) when it is connected to a load is always smaller than the emf $(\mathcal{E})$ and is called the terminal voltage of the battery.



➢ When current flows through circuit, the terminal voltage is given by:

$$\boxed{\boldsymbol{V = \mathcal{E} - Ir}}$$

$$r: internal\ resistance\ of\ battery$$

The voltage across R is:

$$V = IR$$

$$\mathcal{E} - Ir = IR \longrightarrow \mathcal{E} = IR + Ir \longrightarrow \mathcal{E} = I(R + r)$$

The current through the circuit is given by

$$\boxed{\boldsymbol{I = \frac{\mathcal{E}}{R + r}}}$$

Example 4: car battery whose emf is 12.0 V and whose internal resistance is $0.010\ \Omega$. What is the terminal voltage when the current I drawn from the battery is (a) 10.0 A and (b) 100.0 A?

Solution

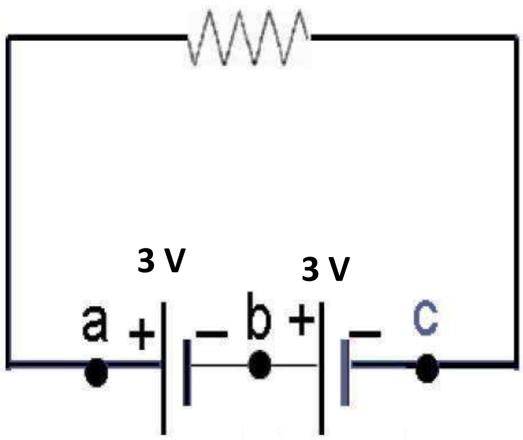
a) $V = \mathcal{E} - Ir = 12 - (10 \times 0.01) = 11.9\ V$

b) $V = \mathcal{E} - Ir = 12 - (100 \times 0.01) = 11\ V$

➢ EMFs in series & parallel:

I- EMFs in series

- It is used to increase the voltage obtained from source



$V_{ab} = V_a - V_b = 3\ V \quad (1)$

$V_{bc} = V_b - V_c = 3\ V \quad (2)$

(1) + (2) gives

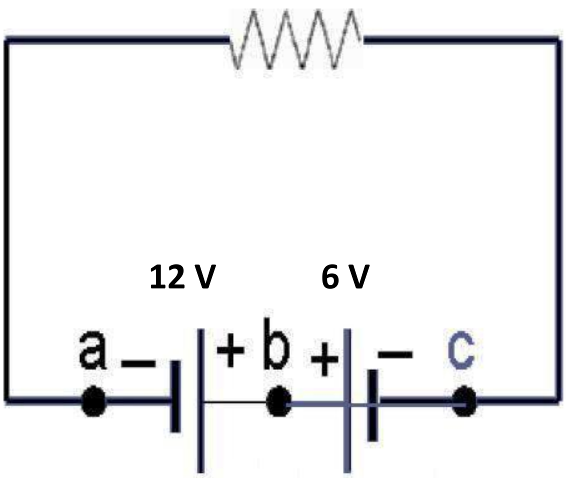
$V_{ac} = V_{ab} + V_{bc} = 6\ V$

$V_{ac} = (V_a - V_b) + (V_b - V_c) = 6\ V$

$V_{ac} = V_a - V_c = 6\ V$

$\therefore V_a > V_c$

So the current will flow from (a) to (c) into the external circuit (through R)



$V_{ab} = V_a - V_b = -12\ V \quad (1)$

$V_{bc} = V_b - V_c = +6\ V \quad (2)$

(1) + (2) gives

$V_{ac} = V_{ab} + V_{bc} = -6\ V$

$V_{ac} = (V_a - V_b) + (V_b - V_c) = -6\ V$

$V_{ac} = V_a - V_c = -6\ V$

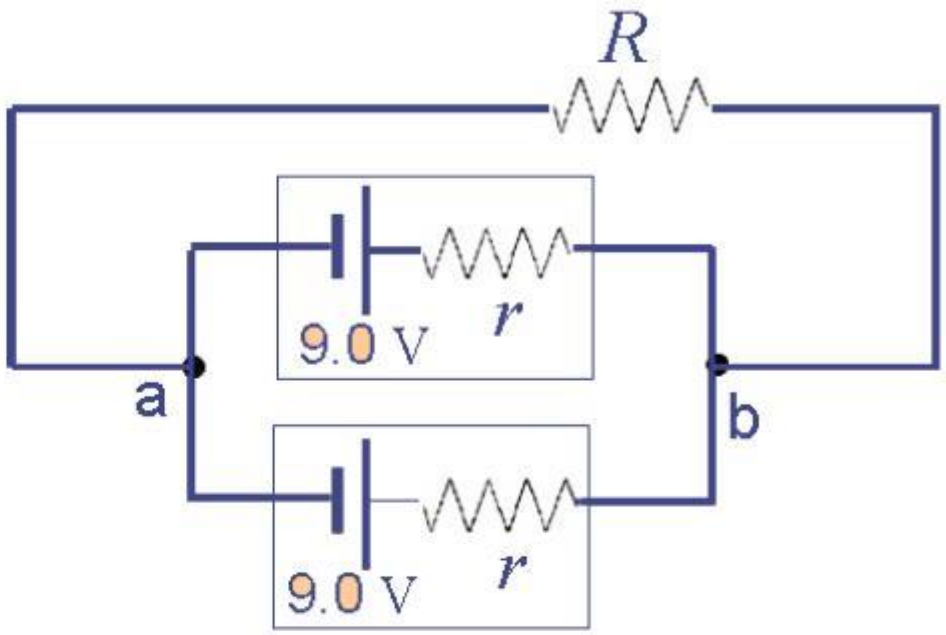
$\therefore V_a < V_c$

So the current will flow from (c) to (a) into the external circuit (through R)

- For batteries in series, the total internal resistance increase.

II- EMFs in parallel

- It is used to increase the current obtained from source

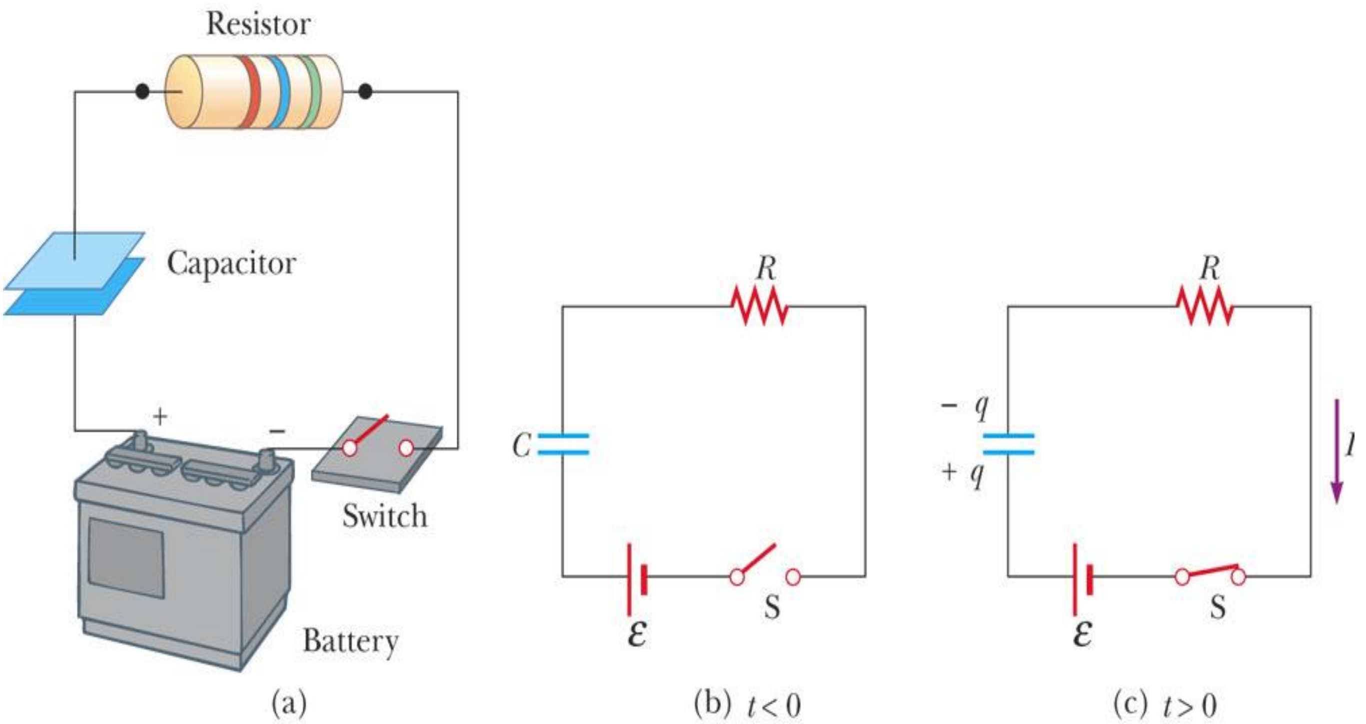


- The voltage is the same as one battery.
- Such an arrangement lasts twice as long as one cell because each of the sources has to produce only a fraction of the total current, and also the loss due to internal resistance is less than for a single cell (i.e. provides more energy).
- For batteries in parallel, the total internal resistance decreases.
- The internal resistance of the two batteries has to be the same, unless the battery with less internal resistance could be damaged due to reverse current.

➢ RC circuits:

1- Charging a capacitor

- It is a circuit consisting of a battery of emf "$\boldsymbol{\varepsilon}$", a resistance "R" & a capacitor "C" connected in series.



- When the switch "S" is closed, current will flow in the circuit and the capacitor will be charged.
- The charge $q(t)$ at any time (t) on the capacitor is given by

$$\boxed{\boldsymbol{q(t) = q_o\left(1 - e^{-t/\tau}\right) = \varepsilon C\left(1 - e^{-t/\tau}\right) = \varepsilon C\left(1 - e^{-t/RC}\right)}}$$

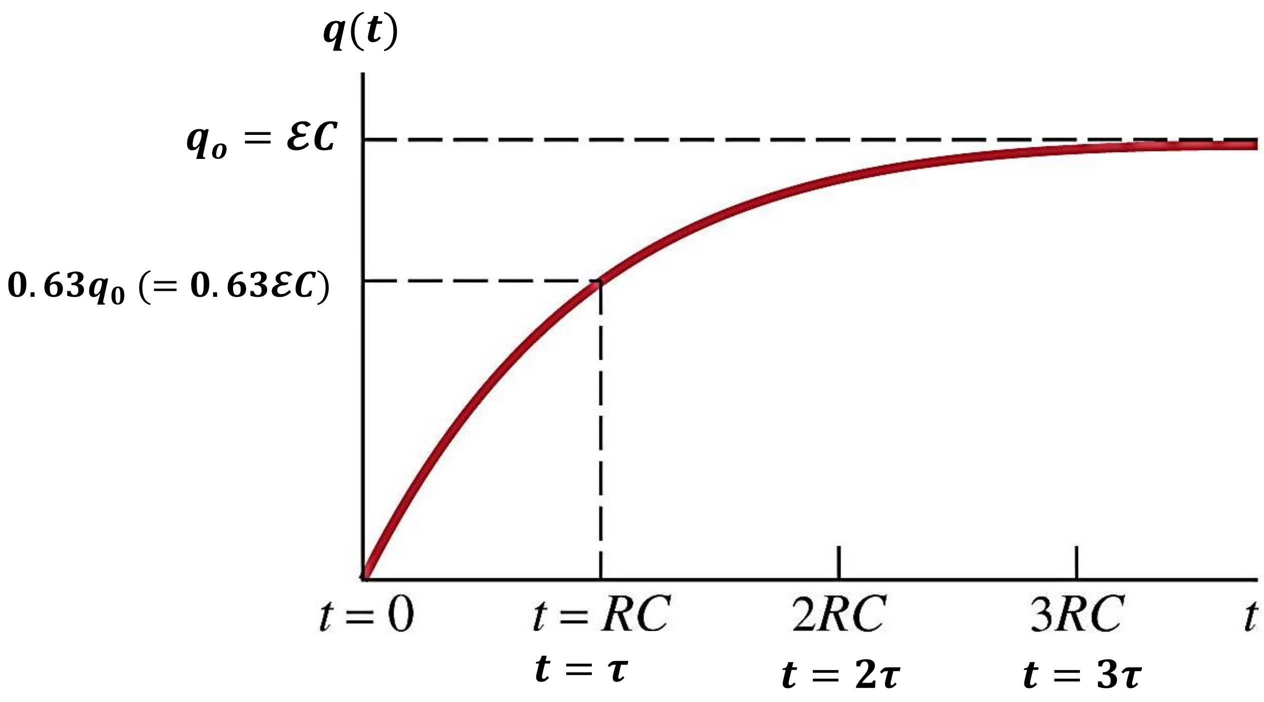
q_o: maximum charge on the capacitor $q_o = \mathcal{E}C$

- The charge will starts from zero till it reaches q_o.

τ: time constant of the circuit $\boxed{\boldsymbol{\tau = RC}}$, it is the time required for the charge on the capacitor to reach (0.63) from its maximum value q_o while charging.

<u>At $t = \tau$:</u>

$$q(\tau) = q_o\left(1 - e^{-\tau/\tau}\right) = q_o(1 - e^{-1}) = q_o(1 - 0.37) = 0.63\ q_o$$



The potential difference between the 2 plates of the capacitor at any time (t) is given by

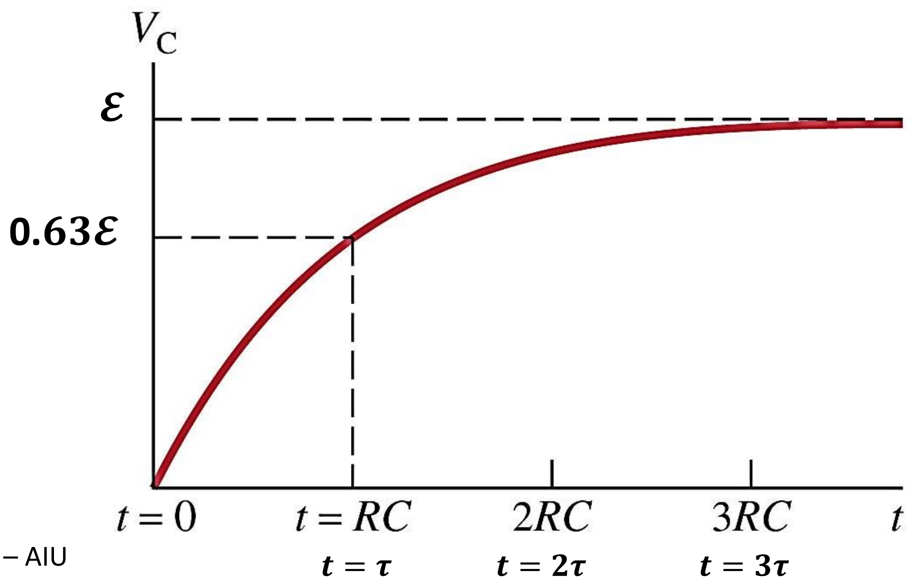
$$V_C(t) = \frac{q(t)}{C} = \frac{q_o}{C}\left(1 - e^{-t/\tau}\right) \rightarrow \boxed{\boldsymbol{V_C(t) = \mathcal{E}\left(1 - e^{-t/\tau}\right)}}$$

V_C starts from zero till it reaches $\mathcal{E}$.

τ: time constant of the circuit $\boxed{\boldsymbol{\tau = RC}}$, it is the time required for the voltage across the capacitor to reach (0.63) from its maximum value $\mathcal{E}$ while charging.

<u>At $t = \tau$:</u>

$$V_C(t) = \mathcal{E}\left(1 - e^{-\tau/\tau}\right) = \mathcal{E}(1 - e^{-1}) = \mathcal{E}(1 - 0.37) = 0.63\ \mathcal{E}$$



The current at any time in the circuit is given by:

$$\boxed{\boldsymbol{I(t) = I_o e^{-t/\tau} = \frac{\mathcal{E}}{R} e^{-t/\tau}}}$$

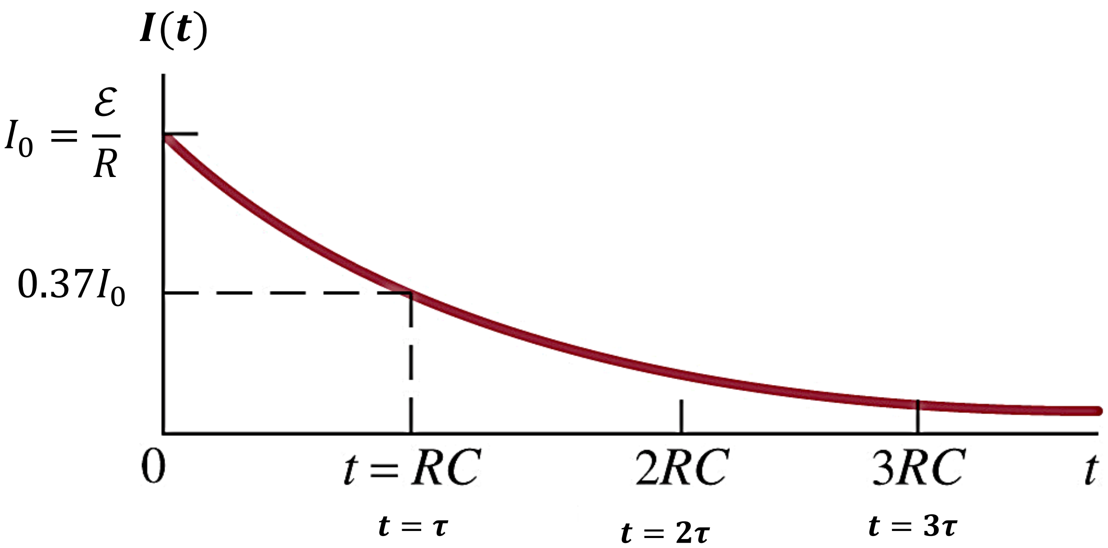
I_o: maximum value of current $I_o = \frac{\mathcal{E}}{R}$

The current starts from I_o and decays to zero.

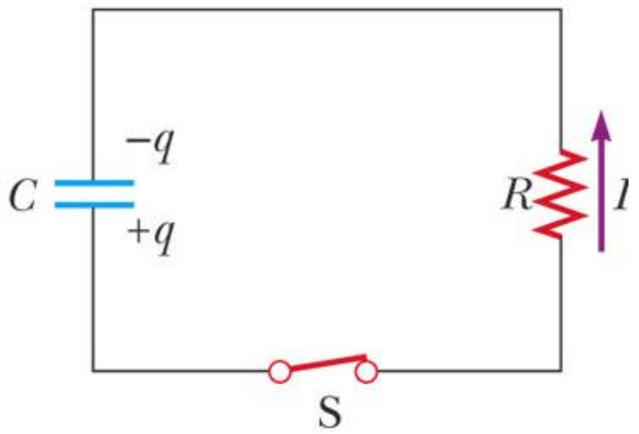
τ: time constant of the circuit $\boxed{\boldsymbol{\tau = RC}}$, it is the time required for the current to reach (0.37) from its maximum value "$I_o = \mathcal{E}/R$" while charging.

<u>At $t = \tau$:</u>

$$I(t) = \frac{\mathcal{E}}{R}\left(e^{-\tau/\tau}\right) = I_o(e^{-1}) = 0.37 I_o$$



2- Discharging a capacitor



- If the battery is removed & the switch is closed, the capacitor will discharges its charge in the resistance.
- The charge $q(t)$ at any time (t) on the capacitor is given by

$$\boxed{\boldsymbol{q(t) = q_o e^{-t/\tau} = \mathcal{E}C\, e^{-t/\tau} = \mathcal{E}C\, e^{-t/RC}}}$$

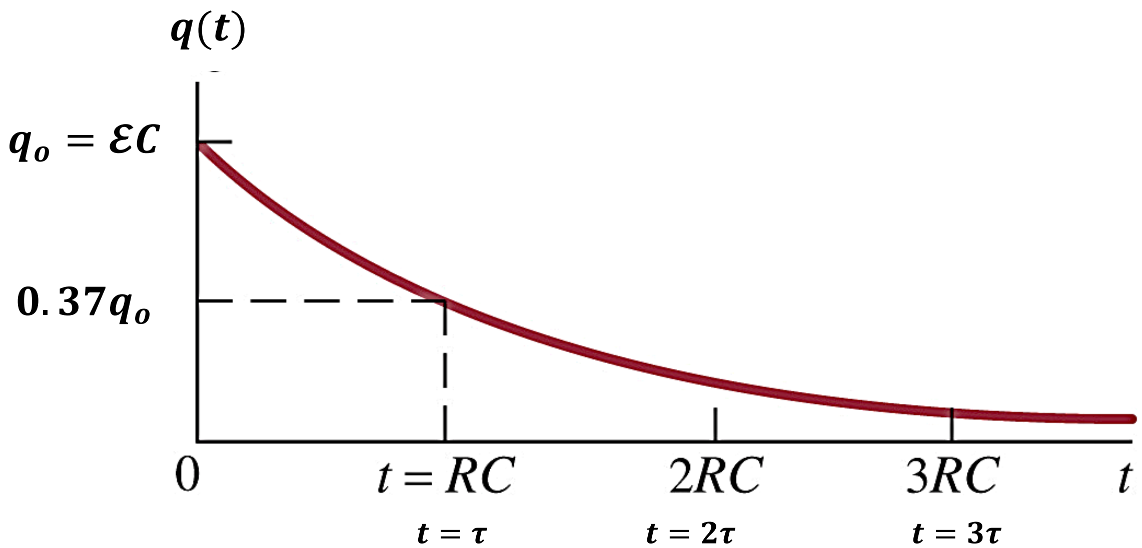
q_o: maximum charge on the capacitor $q_o = \mathcal{E}C$

- The charge will starts from q_o till it reaches zero.

τ: time constant of the circuit $\tau = RC$, it is the time required for the charge on the capacitor to reach (0.37) from its maximum value while discharging.

At $t = \tau$:

$$q(\tau) = q_o\, e^{-\tau/\tau} = q_o\, e^{-1} = 0.37\, q_o$$



- The potential difference between the 2 plates of the capacitor at any time (t) is given by

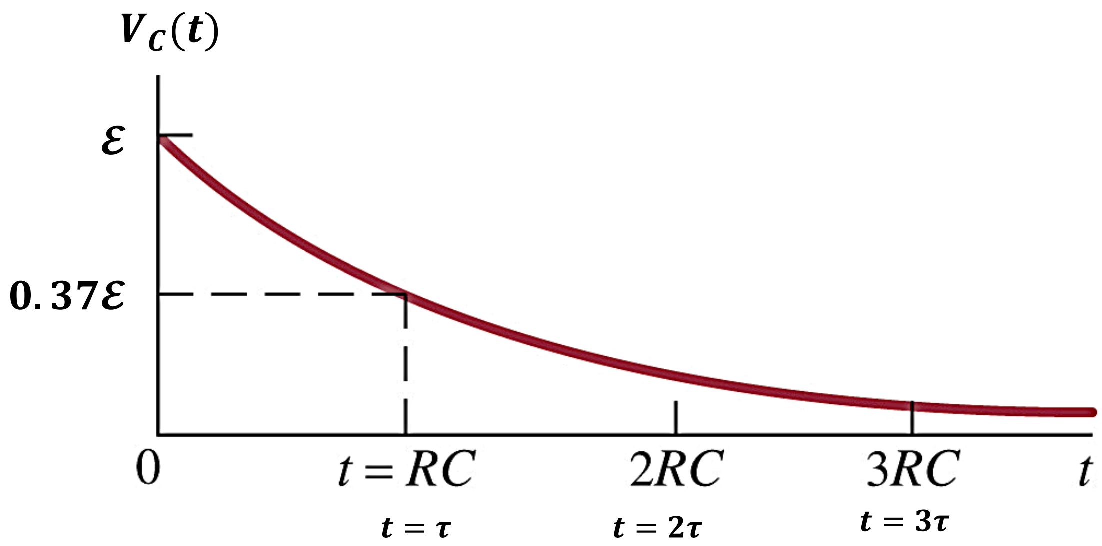
$$V_C(t) = \frac{q(t)}{C} = \frac{q_o}{C} e^{-t/\tau} \rightarrow \boxed{\boldsymbol{V_C(t) = \mathcal{E}\, e^{-t/\tau}}}$$

V_C starts from $\mathcal{E}$ till it reaches zero.

τ: time constant of the circuit $\tau = RC$, it is the time required for the voltage across the capacitor to reach (0.37) from its maximum value while discharging.

<u>At $t = \tau$:</u>

$$V_C(\tau) = \mathcal{E}\, e^{-\tau/\tau} = \mathcal{E}\, e^{-1} = 0.37\, \mathcal{E}$$



- The current at any time in the circuit is given by:

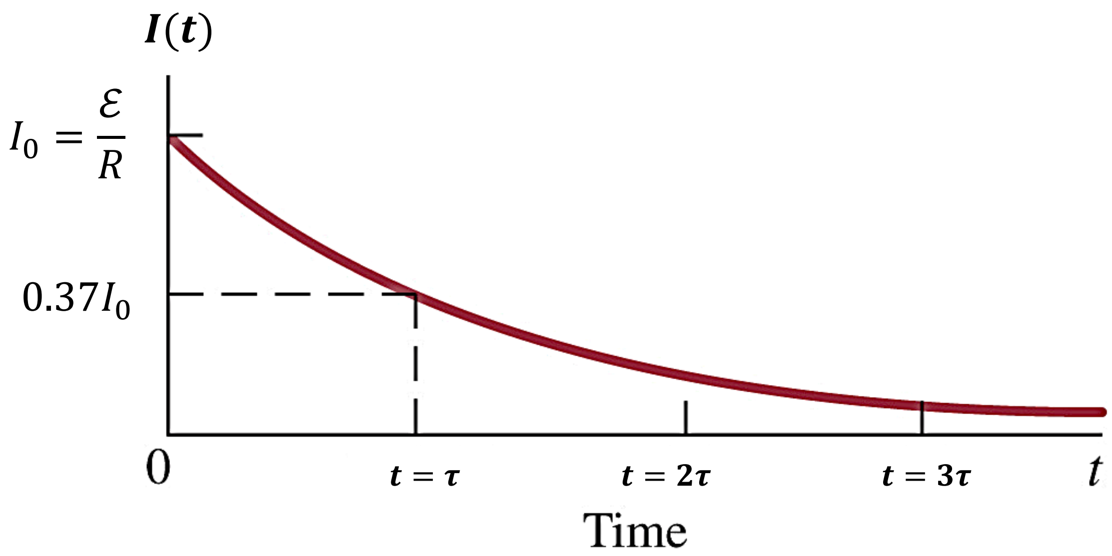
$$\boxed{\boldsymbol{I(t) = I_o e^{-t/\tau} = \frac{\mathcal{E}}{R} e^{-t/\tau}}}$$

The current starts from I_o and decays to zero.

τ: time constant of the circuit $\boxed{\boldsymbol{\tau = RC}}$, it is the time required for the current to reach (0.37) from its maximum value "$I_o = \mathcal{E}/R$" while discharging.

<u>At $t = \tau$:</u>

$$I(t) = \frac{\mathcal{E}}{R}\left(e^{-\tau/\tau}\right) = I_o(e^{-1}) = 0.37 I_o$$



- **Example 5:**

The switch on an RC circuit is closed at $t = 0$. Given that $\mathcal{E} = 6$ V, $R = 150\ \Omega$ & $C = 23\ \mu$F, how much charge is on the capacitor at time $t = 4.2$ ms? & calculate the electric potential energy across the capacitor.

Solution

$$\because q(t) = q_o\left(1 - e^{-t/\tau}\right) = \mathcal{E}C\left(1 - e^{-t/\tau}\right) = \mathcal{E}C\left(1 - e^{-t/RC}\right)$$

$$q(4.2ms) = (6 \times 23 \times 10^{-6})\left(1 - e^{\left(-4.2\times10^{-3}/(150\times23\times10^{-6})\right)}\right) = 97 \times 10^{-6}\ C$$

$$U_c = \frac{qV_C}{2} = q\ \times \mathcal{E}\left(1 - e^{-t/\tau}\right) = \frac{97 \times 10^{-6} \times 6}{2}\left(1 - e^{\left(-4.2\times10^{-3}/(150\times23\times10^{-6})\right)}\right) = 2.05 \times 10^{-4}\ \mathrm{J}$$

- **Example 6:**

Consider an RC circuit with $\mathcal{E} = 12\ V$, $R = 175\ \Omega$ and $C = 55.7\ \mu F$. Find:

a) The time constant of the circuit
b) The maximum charge on the capacitor
c) The initial current in the circuit

Solution

a) $\tau = RC = 175 \times (55.7 \times 10^{-6}) = 9.75 \times 10^{-3}\ sec = 9.75\ ms$
b) $q_o = \mathcal{E}C = 12 \times (55.7 \times 10^{-6}) = 6.68 \times 10^{-4}\ C$
c) $I_o = \frac{\mathcal{E}}{R} = \frac{12}{175} = 0.068\ A$